

Problem 15.16

The cash price of a new automobile is \$10,000. The purchaser is willing to finance the car at 18% convertible monthly and to make payments of \$250 at the end of each month for 4 years. Find the down payment which will be necessary.

Solution

The nominal annual interest rate convertible monthly is 18%, so the monthly interest rate is

$$j = \frac{0.18}{12} = 0.015.$$

There are

$$4(12) = 48$$

monthly payments.

The cash price is equal to the down payment plus the present value of the loan payments. Thus,

$$10000 = \text{Down payment} + 250a_{\overline{48}|j}.$$

Using

$$a_{\overline{n}|j} = \frac{1 - (1 + j)^{-n}}{j},$$

the present value of the monthly payments is

$$\begin{aligned} 250a_{\overline{48}|j} &= 250 \left(\frac{1 - (1.015)^{-48}}{0.015} \right) \\ &\approx 8510.6384. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Down payment} &= 10000 - 8510.6384 \\ &\approx 1489.3616. \end{aligned}$$

Hence, the necessary down payment is

$$\boxed{\$1,489.36}.$$